

OSIM Lab

Full Architecture Build Sheet | Blue/Red Node Design | 2026-03-31

Document	OSIM-ARCH-001	Status	ACTIVE	Version	1.0
Author	Alec Sanchez	Node A	P520 — Blue Team / SOC	Node B	T480 — Red Team / Attack
Repo	github.com/osim-framework/osim-core	Firewall	T620 OPNsense 26.1.5	Switches	Cisco 3560CX + SF300-24P

1. Design Principles

- **Strict blue/red separation:** P520 is blue-only (SOC, detection, threat intel, enterprise targets). T480 is red-only (attack, C2, malware analysis, exploit dev). No red tooling on P520, no blue stack on T480.
- **One-way telemetry:** Wazuh agents and Velociraptor agents on Windows targets send data to P520 SECURITY VLAN only. Red team has no direct access to the blue stack.
- **OPNsense mediates all cross-zone traffic:** Every inter-VLAN path is explicitly permitted. Default deny. All rules logged.
- **Caldera on P520:** Caldera server runs in SECURITY VLAN on P520 so it can reach both TheHive/Wazuh (same node) and Windows targets across a single firewall rule. T480 runs Caldera agents only.
- **Malware sandbox fully isolated:** VLAN 70 (MAL-SANDBOX) has no real WAN access. All outbound traffic redirected to VLAN 80 (FAKE-INTERNET) running INetSim.
- **Cert-aligned:** Every zone and tool maps to at least one certification in the OSCP→OSCE3→GCIA→GCFA→GCTI→GMON→CISSP→SABSA→CCSP path.

2. Hardware Inventory

Machine	CPU	RAM	Role	Proxmox	Current VMs
T620	—	—	OPNsense 26.1.5 firewall/router	No	Bare metal firewall
P520	Xeon W-2135	64GB	Blue node — SOC, detection, targets	Yes	VM100 Wazuh, VM101 TheHive, VM102 MISP
T480	i7-8650U	48GB	Red node — attack, C2, malware analysis	To install	None yet
Cisco 3560CX	—	—	Primary managed switch	No	Trunk to P520, access to T620, MacBook
SF300-24P	—	—	Secondary managed switch	No	Expansion ports
Debian box	—	—	Wiki.js deployed	No	Wiki.js
MacBook M1	M1	—	Management workstation	No	VLAN 11 MGMT-B

3. VLAN Architecture

Existing VLANs (10, 11, 20, 40) are unchanged. New VLANs (30, 50, 60, 70, 80, 90, 91, 92) are added to OPNsense and the Cisco 3560CX trunk.

VLAN	Name	Subnet	Gateway	Purpose	Hosts	Status
10	MGMT-A	10.10.10.0/24	10.10.10.1	Proxmox/infra admin	P520, T480 mgmt interfaces	Existing
11	MGMT-B	10.10.11.0/24	10.10.11.1	MacBook management workstation	MacBook, Proxmox web UI	Existing
20	LAB-OPS	10.10.20.0/24	10.10.20.1	TheHive, MISP, blue supporting apps	VM101, VM102	Existing
30	WINDOWS-ENT	10.10.30.0/24	10.10.30.1	AD DC, workstations, app/file server	Win DC, Win clients, IIS/SQL	NEW
40	SECURITY	10.10.40.0/24	10.10.40.1	Wazuh, Velociraptor, Greenbone, Caldera	VM100, Velociraptor, Caldera	Existing
50	RED-OPS	10.10.50.0/24	10.10.50.1	T480 attacker tooling — Kali, BloodHound	T480 Kali VM	NEW
60	C2-ISOLATED	10.10.60.0/24	10.10.60.1	Sliver/Havoc listeners, C2 beacon sim	T480 Sliver VM	NEW
70	MAL-SANDBOX	10.10.70.0/24	10.10.70.1	Malware detonation — no real WAN	T480 CAPE, REMnux	NEW
80	FAKE-INTERNET	10.10.80.0/24	10.10.80.1	INetSim fake internet — DNS/HTTP/SMTP	T480 INetSim VM	NEW
90	OSWE-WEB	10.10.90.0/24	10.10.90.1	Web app targets for OSWE practice	P520 DVWA/WebGoat VMs	NEW
91	OSED-TARGET	10.10.91.0/24	10.10.91.1	Exploit dev Windows targets for OSED	T480 Win10 x86/x64 VMs	NEW
92	JUMP-ADMIN	10.10.92.0/24	10.10.92.1	Hardened jump host for Windows admin/RDP	P520 Debian jump VM	NEW

4. P520 Blue Node — VM Inventory (64GB RAM)

Total allocated: ~52GB. Proxmox overhead: 4GB. Headroom: ~8GB for snapshots and memory ballooning. Power off Windows workstations during heavy blue-team analysis sessions.

VMID	Name	OS	RAM	vCPU	IP	VLAN	Role / Tools
100	wazuh-manager	Ubuntu 22.04	16GB	4	10.10.40.10	40	Wazuh Manager + Indexer + Dashboard. Primary SIEM. Receives agents from VLAN 30 and 40.
101	thehive	Ubuntu 22.04	4GB	2	10.10.20.10	20	TheHive 5.4 (Docker). Case management. Receives alerts from Wazuh via API bridge.
102	misp	Ubuntu 22.04	4GB	2	10.10.20.11	20	MISP 2.4. Threat intelligence platform. IOC enrichment for TheHive cases.
103	greenbone	Greenbone OS	4GB	2	10.10.40.20	40	Greenbone/OpenVAS. Vulnerability scanning of Windows targets in VLAN 30. Feeds vuln context to TheHive.
104	velociraptor	Ubuntu 22.04	2GB	2	10.10.40.30	40	Velociraptor server. Live response and artifact collection from Windows targets. Critical for GCFA.
105	caldera	Ubuntu 22.04	2GB	2	10.10.40.40	40	Caldera server. Runs ATT&CK-mapped; adversary simulations against VLAN 30 targets. T480 runs agents only.
106	win-dc01	Windows Server 2022	4GB	2	10.10.30.10	30	AD Domain Controller. Domain: osim.lab. OUs, GPOs, users, service accounts. Wazuh + Velociraptor agents.
107	win-client01	Windows 11	4GB	2	10.10.30.20	30	Domain-joined workstation. Sysmon + Wazuh + Velociraptor agents. Primary attack target for OSCP/OSEP.
108	win-client02	Windows 11	4GB	2	10.10.30.21	30	Second domain-joined workstation. Used for lateral movement simulation and AD attack chain practice.
109	win-server01	Windows Server 2022	4GB	2	10.10.30.30	30	Member server. IIS, MSSQL Express, SMB shares, WinRM. Realistic enterprise attack surface.
110	jump-admin	Debian 12	2GB	1	10.10.92.10	92	Jump host. RSAT, RDP client, AD management tools. MacBook never touches VLAN 30 directly.
111	oswe-web	Ubuntu 22.04	2GB	1	10.10.90.10	90	DVWA + WebGoat + custom IIS app. Web app targets for OSWE practice. Reachable from RED-OPS only.

P520 RAM summary: Wazuh 16GB + TheHive 4GB + MISP 4GB + Greenbone 4GB + Velociraptor 2GB + Caldera 2GB + Win DC 4GB + Win Client x2 8GB + Win Server 4GB + Jump 2GB + OSWE 2GB = 52GB allocated. Proxmox overhead 4GB. Total 56GB. Headroom 8GB.

5. T480 Red Node — VM Inventory (48GB RAM)

Total allocated: ~40GB. Proxmox overhead: 4GB. Headroom: ~4GB. Run CAPE and FlareVM one at a time, never simultaneously. OSED target VMs are snapshots — power off when not in use.

VMID	Name	OS	RAM	vCPU	IP	VLAN	Role / Tools
200	kali-purple	Kali Linux 2024	8GB	4	10.10.50.10	50	Primary attacker VM. Full Kali Purple toolset. Metasploit, Impacket, netexec, CrackMapExec, BloodHound ingestors, Burp Suite. OSCP/OSEP primary platform.
201	sliver-c2	Ubuntu 22.04	4GB	2	10.10.60.10	60	Sliver C2 server. Isolated in C2-ISOLATED VLAN. Listeners receive beacons from Windows targets. Havoc C2 also deployed here for OSEP evasion comparison.
202	remnux	REMnux 7	8GB	4	10.10.70.10	70	REMnux full VM. Malware analysis: strings, YARA, pefile, Volatility3, Cutter, Ghidra, radare2, oledump. GCFA malware analysis workflows.
203	flarevm	Windows 10	8GB	4	10.10.70.20	70	FlareVM Windows malware analysis platform. x64dbg, WinDbg, IDA Free, PE-bear, PESTudio, dnSpy. Do not run simultaneously with CAPE.
204	cape-sandbox	Ubuntu 22.04	8GB	4	10.10.70.30	70	CAPE sandbox. Automated malware detonation. All outbound traffic routed to VLAN 80 INetSim. No real WAN access. Run independently of FlareVM.
205	inetsim	Debian 12	2GB	1	10.10.80.10	80	INetSim fake internet gateway. Simulates DNS, HTTP, HTTPS, SMTP, FTP. All VLAN 70 and VLAN 60 outbound traffic redirected here by OPNsense.
206	osed-win10-x86	Windows 10 x86	4GB	2	10.10.91.10	91	OSED exploit development target. Windows 10 x86 with specific patch levels for buffer overflow and SEH chain practice. WinDbg + Immunity Debugger.
207	osint-vm	Ubuntu 22.04	4GB	2	10.10.50.20	50	OSINT workstation. Maltego Community, Spiderfoot, Recon-ng, theHarvester, Shodan CLI, OSINT framework tools. GCTI recon workflows.

T480 RAM summary: Kali 8GB + Sliver 4GB + REMnux 8GB + FlareVM 8GB + CAPE 8GB + INetSim 2GB + OSED Win10 4GB + OSINT 4GB = 46GB. But FlareVM + CAPE + OSED = 20GB — never run all three simultaneously. Realistic working set with two heavy VMs: ~34GB. Proxmox overhead 4GB. Total 38GB. Headroom 10GB when one heavy VM is suspended.

6. OPNsense Firewall Rule Matrix

All rules are PASS unless marked BLOCK. Default policy on all interfaces is BLOCK. Rules are evaluated top-down first-match. All PASS rules should have logging enabled.

Rule ID	Source VLAN	Dest VLAN	Protocol	Ports	Action	Purpose / Cert Relevance
FW-01	WINDOWS-ENT (30)	SECURITY (40)	TCP/UDP	1514, 1515, 55000	PASS	Wazuh agent enrollment and event forwarding from Windows targets to Wazuh manager. Critical for all GCIA/GMON telemetry.
FW-02	WINDOWS-ENT (30)	SECURITY (40)	TCP	8000, 8001	PASS	Velociraptor agent communication from Windows targets to Velociraptor server on P520. GCFA live response.
FW-03	SECURITY (40)	WINDOWS-ENT (30)	TCP	Any	PASS	Greenbone/OpenVAS scanner reaching Windows targets. Vuln scan for vulnerability management workflow.
FW-04	WINDOWS-ENT (30)	SECURITY (40)	TCP	8888	PASS	Caldera agent beaconing back to Caldera server on P520. Required for ATT&CK; simulation operations.
FW-05	RED-OPS (50)	WINDOWS-ENT (30)	Any	Any	PASS	T480 Kali attacking Windows enterprise targets. Full access for OSCP/OSEP practice. Log all traffic.
FW-06	RED-OPS (50)	OSWE-WEB (90)	TCP	80, 443, 8080, 8443	PASS	T480 attacking web application targets. Burp Suite, SQLmap, web exploit chains for OSWE practice.
FW-07	RED-OPS (50)	C2-ISOLATED (60)	TCP	Any	PASS	T480 Kali connecting to Sliver/Havoc C2 server for payload staging and operator console access.
FW-08	WINDOWS-ENT (30)	C2-ISOLATED (60)	TCP	443, 80, 8443	PASS	Simulated C2 beacon traffic from compromised Windows targets back to Sliver listener. OSEP evasion practice.
FW-09	C2-ISOLATED (60)	FAKE-INTERNET (80)	Any	Any	PASS	C2 VLAN outbound redirected to INetSim fake internet. Simulates realistic egress without real WAN exposure.
FW-10	MAL-SANDBOX (70)	FAKE-INTERNET (80)	Any	Any	PASS	All malware sandbox outbound redirected to INetSim. CAPE detonation gets fake DNS/HTTP responses.
FW-11	MAL-SANDBOX (70)	WAN	Any	Any	BLOCK	Hard block — malware sandbox has zero real WAN access. INetSim is the only permitted outbound target.
FW-12	C2-ISOLATED (60)	WAN	Any	Any	BLOCK	Hard block — C2 VLAN cannot reach real internet. All simulated egress goes to FAKE-INTERNET only.
FW-13	RED-OPS (50)	SECURITY (40)	Any	Any	BLOCK	Red team cannot directly access Wazuh, TheHive, MISP, or Caldera server. Preserves blue stack integrity.

FW-14	RED-OPS (50)	LAB-OPS (20)	Any	Any	BLOCK	Red team cannot access TheHive or MISP directly. No threat intel exfil or case manipulation from red node.
FW-15	RED-OPS (50)	OSD-TARGET (91)	Any	Any	PASS	T480 reaching OSD exploit development target VMs. Controlled access for buffer overflow and OSD practice.
FW-16	MGMT-B (11)	All VLANs	TCP	22, 443, 8006, 9000	PASS	MacBook management access to all systems — SSH, Proxmox web UI (8006), TheHive (9000), OPNsense (443).
FW-17	JUMP-ADMIN (92)	WINDOWS-ENT (30)	TCP	3389, 445, 135, 389	PASS	Jump host RDP and AD management tools reaching Windows targets. MacBook RDPs to jump host, never directly to VLAN 30.

7. Deployment Order — Phased Build Plan

Each phase has a validation gate. Do not proceed to the next phase until the gate passes. This prevents building offensive tooling before detection is confirmed working.

Phase	Step	Task	Validation Gate	Est. Time
PHASE 1 Network	1.1	Add VLANs 30,50,60,70,80,90,91,92 to OPNsense interfaces and Cisco 3560CX trunk (Gi0/6)	ping between MGMT-B and each new gateway IP from OPNsense shell	30 min
	1.2	Create OPNsense aliases for new subnets (WINDOWS-ENT, RED-OPS, C2-ISOLATED, MAL-SANDBOX, FAKE-INTERNET, OSWE-WEB, OSED-TARGET, JUMP-ADMIN)	pfctl -t WINDOWS-ENT -T show returns 10.10.30.0/24	20 min
	1.3	Implement firewall rules FW-01 through FW-17 on OPNsense. Verify FW-11, FW-12, FW-13, FW-14 (block rules) are active before any red tooling is deployed	From RED-OPS test IP, confirm cannot reach 10.10.40.x or 10.10.20.x	45 min
PHASE 2 Windows Targets	2.1	Deploy Windows Server 2022 DC (VM106) on P520. Install Proxmox VM, assign VLAN 30 tag in VM config. Configure AD domain: osim.lab. Create OUs, GPOs, 10 test users, 2 service accounts, 1 helpdesk account	dcdiag /test:connectivity passes. Domain reachable from jump host VM.	2 hrs
	2.2	Install Sysmon on DC using Olaf Hartong modular config. Install Wazuh agent, enroll to 10.10.40.10. Install Velociraptor agent pointing to 10.10.40.30	Wazuh dashboard shows DC agent active. Velociraptor server shows DC enrolled. Sysmon events visible in Wazuh.	45 min
	2.3	Deploy Windows 11 client x2 (VM107, VM108). Join domain. Install Sysmon + Wazuh + Velociraptor agents on both	All 3 Windows agents show active in Wazuh. Velociraptor shows 3 clients.	1.5 hrs
	2.4	Deploy Windows member server (VM109). Install IIS, MSSQL Express, configure SMB shares, enable RDP and WinRM. Join domain. Install agents	IIS default page reachable from jump host. MSSQL service running. Wazuh agent active.	1 hr
PHASE 3 Blue Stack Expansion	3.1	Deploy Greenbone VM (VM103) on P520 VLAN 40. Configure scan target list for 10.10.30.0/24. Run first authenticated scan against Windows DC	Scan completes, vulnerabilities visible in Greenbone dashboard. FW-03 confirmed working.	1 hr
	3.2	Deploy Velociraptor server VM (VM104) on P520 VLAN 40. Configure agent MSI for Windows targets. Re-enroll all Windows VMs	All 4 Windows VMs show enrolled in Velociraptor. Run a basic artifact hunt (Windows.System.Pslist) and confirm results	45 min

	3.3	Deploy Caldera server VM (VM105) on P520 VLAN 40. Configure agent profile for Windows targets. Install first Caldera agent on win-client01	Caldera agent beacons from win-client01. Run Discovery operation, confirm Wazuh alerts fire, TheHive case created.	1 hr
	3.4	Run first Atomic Red Team test suite on win-client01 (T1059 Command and Scripting). Verify detection pipeline end to end	Wazuh alert fires → TheHive case auto-created → MISP enrichment available. Full pipeline confirmed before red tools deployed.	45 min
PHASE 4 Red Node	4.1	Install Proxmox VE on T480. Configure network: nic0 tagged trunk to Cisco switch port for VLANs 50,60,70,80,91. Verify bridge-vlan-aware config from P520 lessons learned	T480 Proxmox web UI reachable at 10.10.10.x from MGMT-A. Bridge VLAN show confirms correct PVID assignments.	1 hr
	4.2	Deploy Kali Purple VM (VM200) on T480 VLAN 50. Install BloodHound CE, Impacket, netexec, Rubeus tools, Burp Suite. Run BloodHound ingestor against osim.lab domain	BloodHound graph shows AD relationships. Kali can reach 10.10.30.x Windows targets. FW-05 confirmed.	1 hr
	4.3	Deploy Sliver C2 VM (VM201) on T480 VLAN 60. Configure HTTP/HTTPS listeners. Generate test implant, deploy on win-client01, confirm beacon	Sliver console shows active session from win-client01. Wazuh detects C2 beacon. TheHive case created.	45 min
	4.4	Deploy REMnux VM (VM202) and INetSim VM (VM205) on T480. Configure OPNsense to redirect all VLAN 70 outbound to 10.10.80.10. Test INetSim DNS interception	nslookup evil.com from VLAN 70 VM returns INetSim response. No real DNS leaks.	1 hr
	4.5	Deploy CAPE sandbox VM (VM204) on T480 VLAN 70. Submit a known-safe test sample (EICAR). Confirm detonation report generated	CAPE analysis report generated. Network connections show INetSim responses only. No real WAN traffic.	1.5 hrs

8. Certification Alignment Matrix

Each certification maps to specific lab components. Use this to prioritize which VMs to build first based on your active cert prep.

Certification	Primary Lab Components	Key Exercises	Phase Ready
OSCP	Kali Purple (VM200), Windows targets (VM106-109), Metasploit, BloodHound	Service enumeration on IIS/MSSQL/SMB. Privilege escalation on Windows clients. AD exploitation chain DC→workstation.	Phase 4.2
OSEP	Sliver C2 (VM201), Kali Purple (VM200), Windows targets, AV evasion tooling	C2 beacon evasion through Windows Defender. Process injection. AMSI bypass. Lateral movement via Sliver sessions.	Phase 4.3
OSWE	OSWE-WEB targets (VM111, VLAN 90), Burp Suite, DVWA, WebGoat, custom IIS app	Source code review on web app targets. Auth bypass chains. Second-order SQLi. Deserialization exploits.	Phase 4.2
OSED	OSED-TARGET Win10 x86 (VM206), WinDbg, x64dbg, Immunity Debugger, pwndbg	Stack buffer overflows. SEH chain exploitation. egghunter shellcode. Windows x86 exploit development workflow.	Phase 4.4
OSCE3	All OSEP + OSWE + OSED components combined	Full chain: initial access via web vuln → AV evasion → C2 beacon → lateral movement → AD takeover.	Phase 4.5
GCIA	Wazuh (VM100), Suricata on OPNsense, Zeek on P520, Windows targets generating traffic	IDS rule writing in Suricata. Network traffic analysis of ATT&CK; simulations. Alert triage in Wazuh dashboard.	Phase 3.4
GCFA	Velociraptor (VM104), REMnux (VM202), FlareVM (VM203), CAPE (VM204), Wazuh	Artifact collection from compromised Windows VM via Velociraptor. Memory forensics with Volatility3 in REMnux. Malware triage in CAPE.	Phase 4.4
GCTI	MISP (VM102), OSINT VM (VM207), OpenCTI (add to P520), Wazuh IOC feeds	Threat actor profiling. IOC enrichment workflow. MISP event creation from CAPE sandbox reports. YARA rule development.	Phase 3.4
GMON	Wazuh (VM100), Caldera (VM105), Greenbone (VM103), Velociraptor (VM104), Sysmon	Continuous monitoring dashboard setup. Vulnerability management workflow. Detection rule tuning after Caldera simulations.	Phase 3.3
CISSP	Full lab architecture — zone model, trust boundaries, access controls, audit logging	Document security architecture using OSIM Framework. Risk assessment of lab trust boundaries. Access control matrix review.	Phase 1
SABSA	OSIM Framework documentation, Wiki.js, zone model, firewall rule matrix	Map lab to SABSA contextual/conceptual/logical/physical layers. Business attribute profiling. Security service catalogue.	Phase 1
CCSP	Add LocalStack or MinIO to Debian box for cloud storage simulation	Cloud security controls mapping. Data classification in simulated cloud storage. IAM policy review. CASB concept practice.	Post Phase 4

9. Tools NOT Recommended

The following tools were considered and rejected for this specific stack and goals:

Tool	Why Not Recommended
Elastic Security	Redundant with Wazuh for your goals. Adds significant RAM overhead on P520 (8-16GB). Wazuh already provides SIEM, EDR alerting, and dashboard. Only add if you specifically want platform comparison for a job requirement.
Splunk Free	500MB/day ingest limit makes it useless for realistic lab telemetry. The paid version is cost-prohibitive. Wazuh provides equivalent capability for your cert path.
Covenant C2	Effectively unmaintained since 2022. Sliver and Havoc are the current standards for OSEP-level C2 practice. No reason to add Covenant.
OpenVAS standalone	Greenbone Community Edition is the maintained fork of OpenVAS and includes the full GSA web interface. Use Greenbone, not bare OpenVAS.
Cuckoo Sandbox	Cuckoo 2.x is abandoned. CAPE sandbox is the actively maintained fork with better Windows 10 support and more analysis modules. Use CAPE.
Maltego paid tier	Maltego Community Edition is sufficient for GCTI lab work. The paid tier adds transforms that rely on external APIs — not useful in an air-gapped lab. Use Spiderfoot HX or Recon-ng for deeper OSINT automation.
Full OpenCTI deployment	OpenCTI is RAM-heavy (8-12GB for a production instance). For your P520 load, MISP covers most GCTI exam scenarios. Add OpenCTI only after all Phase 4 VMs are stable and you have confirmed RAM headroom.